

early, then I will wake up feeling refreshed" lead to the conclusion
 "If I do not finish ~~the~~ writing the program, then I will wake up feeling refreshed."

Solution: p: You send me an email message q: I will finish writing the program
 r: I will go to sleep early s: I will wake up feeling refreshed

Premises: 1) $p \rightarrow q$ 2) $\neg p \rightarrow r$ 3) $p \rightarrow s$

Conclusion: $\neg q \rightarrow \neg s$

Step	Reason
1) $\neg p \rightarrow r$	Premise
2) $p \rightarrow s$	Premise
3) $\neg p \rightarrow \neg s$	Hypothetical syllogism on (1) and (2)
4) $p \rightarrow q$	Premise
5) $\neg q \rightarrow \neg p$	Contrapositive on (4)
6) $\neg q \rightarrow \neg s$	Hypothetical syllogism on (5), (3)

Example 8: Show that the following argument is valid. If today is Tuesday, I have a test of Mathematics or Economics. If my Economics Professor is sick, I will not have a test of Economics. Today is Tuesday and my Economics Professor is sick. Therefore I have a test of Mathematics.

Solution: p: Today is Tuesday q: I have a test of Mathematics
 r: I have a test of Economics s: My Economics Professor is sick

Premises: 1) $p \rightarrow (q \vee r)$ 2) $s \rightarrow \neg r$ 3) $p \wedge s$

Conclusion: q